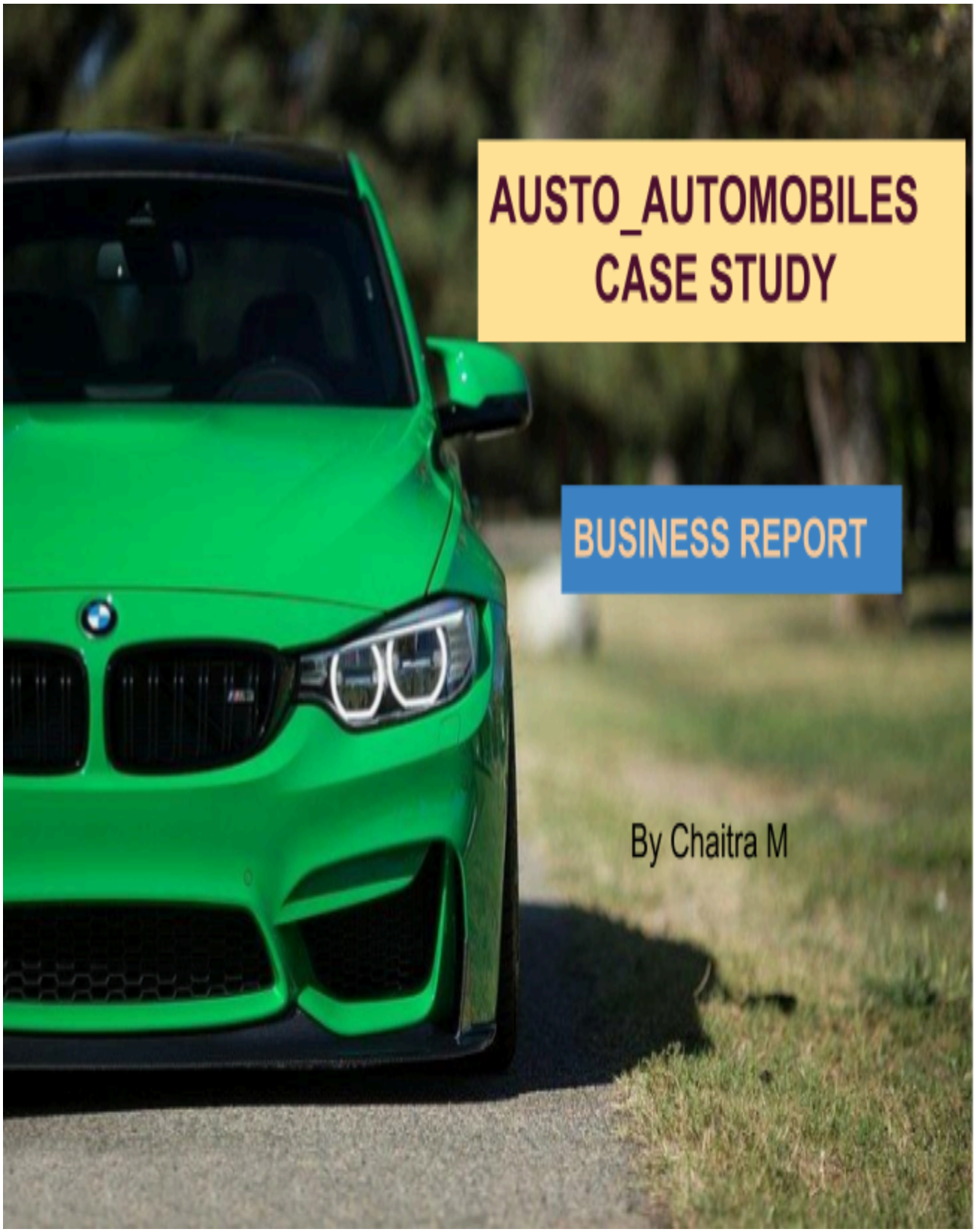

AUSTO_AUTOMOBILES CASE STUDY

BUSINESS REPORT

By Chaitra M

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Background

- Austo Motor Company is a leading car manufacturer specializing in SUV, Sedan, and Hatchback models.
- In its recent board meeting, concerns were raised by the members on the efficiency of the marketing campaign currently being used.
- The board decides to rope in an analytics professional to improve the existing campaign.

Objective

- They want to analyze the data to get a fair idea about the demand of customers which will help them in enhancing their customer experience.

  Our focus will be on:

  - Gender preference for SUVs: Men vs. Women.
  - Likelihood of salaried individuals purchasing sedans.
  - Evidence for Sheldon's claim on salaried males and SUV sales.
  - Gender-based spending on automobiles.
  - Spending by individuals with personal loans.
  - Influence of working partners on high-priced car purchase

Data Description

Variable	Description
Age	The age of the individual in years.
Gender	The gender of the individual, categorized as male or female.
Profession	The occupation or profession of the individual.
Marital_status	The marital status of the individual, such as married & single.
Education	The educational qualification of the individual Graduate and Post Graduate.
No_of_Dependents	The number of dependents (e.g., children, elderly parents) that the individual supports financially.
Personal_loan	A binary variable indicating whether the individual has taken a personal loan "Yes" or "No".
House_loan	A binary variable indicating whether the individual has taken a housing loan "Yes" or "No".
Partner_working	A binary variable indicating whether the individual's partner is employed "Yes" or "No".

Salary	The individual's salary or income.
Partner_salary	The salary or income of the individual's partner, if applicable.
Total_salary	The total combined salary of the individual and their partner (if applicable).
Price	The price of a product or service.
Make	The type of automobile

Table 1: Data Description

Note:

1. There are in total 1581 observations and 14 variables.
2. There are no repeated values, although there are 53 instances where gender information is missing and 106 instances where partner salary data is absent.
3. There are two incorrect entries for gender, namely "Femal" and "Femle“
4. "Female" has been substituted for the incorrect entries "Femal" and "Femle".
5. All null fields in the "Gender" column have been updated to "Unknown".
6. The missing data in partner salary is filled with the difference between Total_salary and Salary.
7. Converted Partner_salary from float to int data type.
8. Created Gender_Male column for plotting the graph.

Univariate Analysis

Understanding the distribution of "Age"

- The graph is right skewed, indicating that older individuals are less.
- The largest number of people are of age 25-35 years old.
- As age goes above 40, the number of people starts to drop.

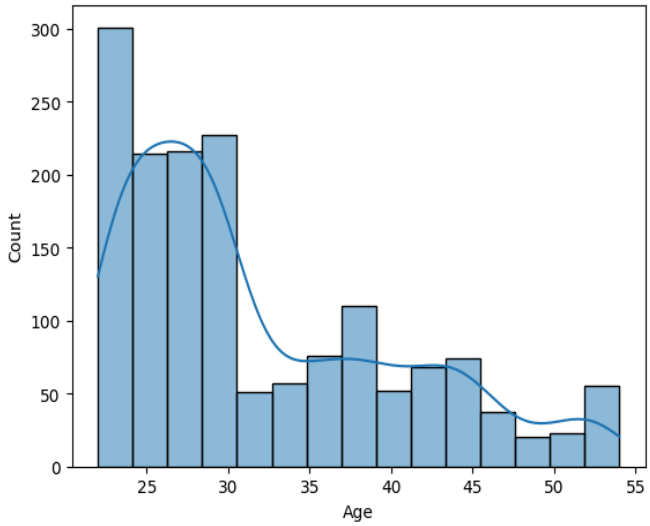

Figure 1: Histogram of Age

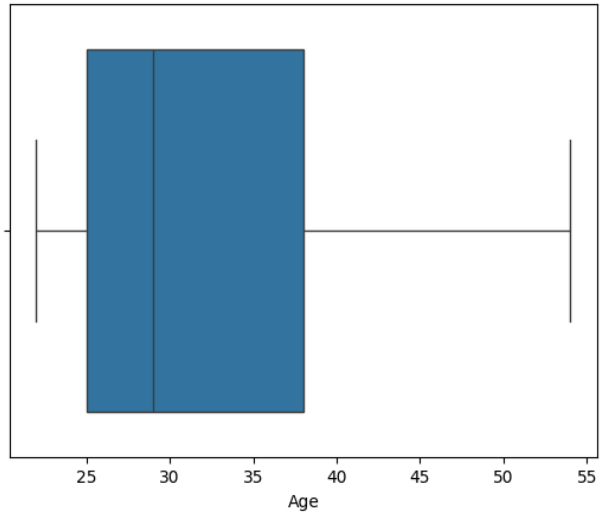

Figure 2: Barplot of Age

Understanding the distribution of Gender

- The number of Male is more as compared to females.

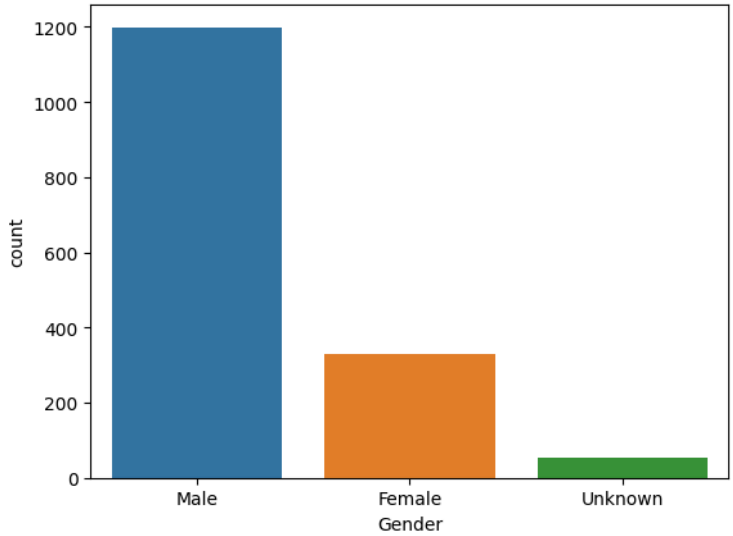

Figure 3: Countplot of Age

Understanding the distribution of Profession

In terms of Profession the salaried individual is more comparatively the one who runs business.

Figure 4: Histogram of Profession

Understanding the distribution of No_of_Dependents

- Majority of the group have 2 or 3 dependencies.
- There are relatively fewer people with 0 or 4 dependency.

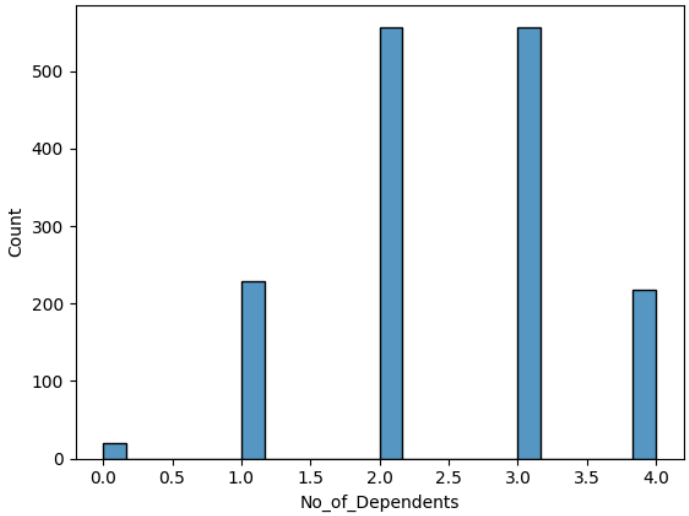

Figure 5: Histogram of No_of Dependents

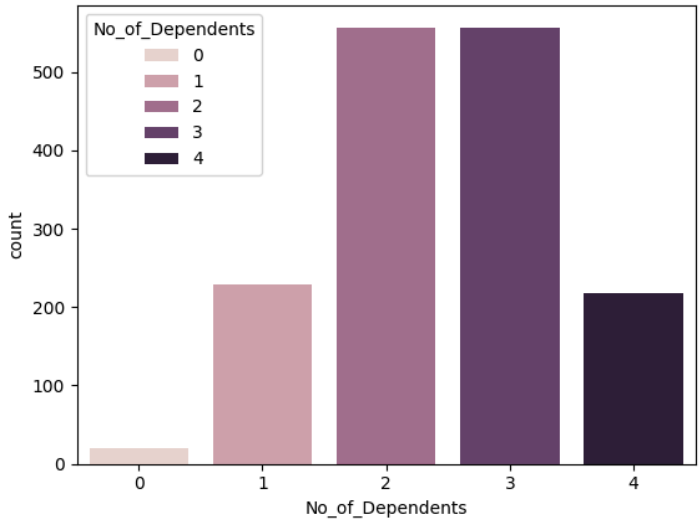

Figure 6: Countplot of No_of Dependents

Understanding the distribution of Salary

- The Salary has normal distribution with slightly right-skewed.
- The majority earn between 55000 to 75000, the small right skew tells that few earn high.
- The median value is 60000.

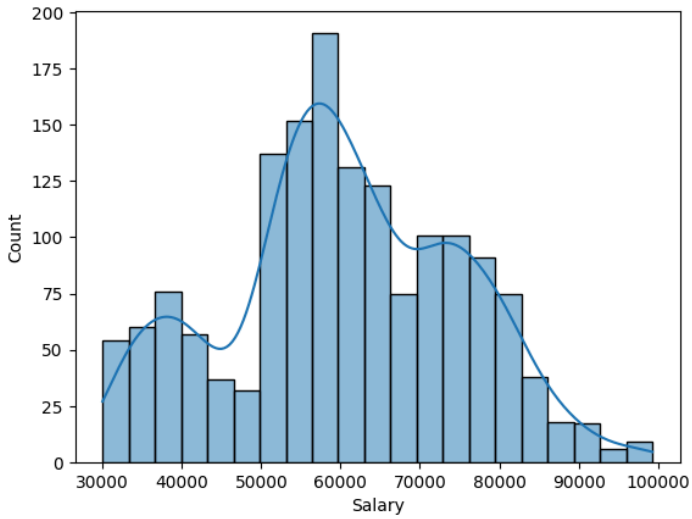

Figure 7: Histogram of Salary

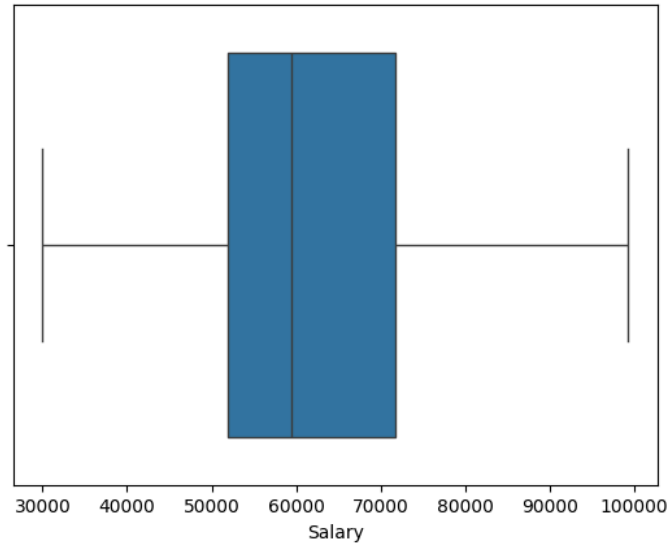

Figure 8: Countplot of Salary

Understanding the distribution of 'Partner_salary

- The graph is highly right skewed. The majority of the salaries are on the lower end.
- The median value of Partner_salary is 25000.

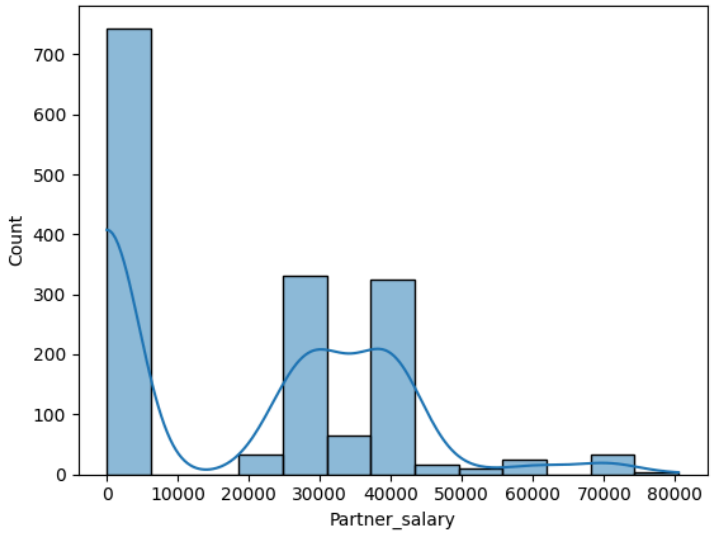

Figure 9: Histogram of Partner_salary

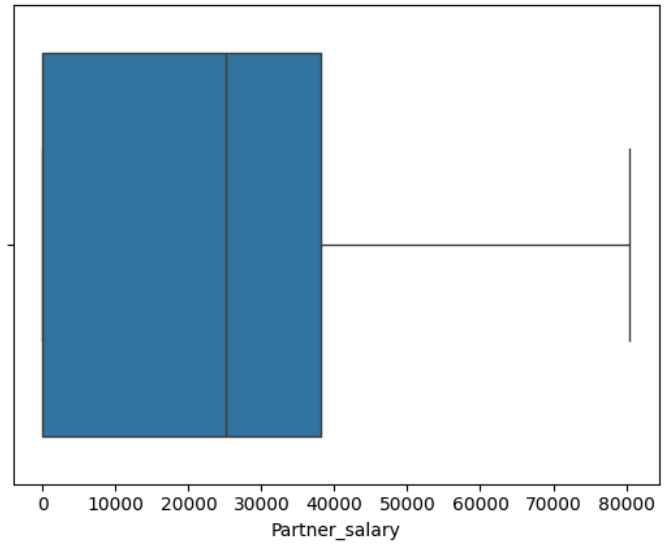

Figure 10: Barplot of Partner_salary

Understanding the distribution of Total_salary

- The "Total_salary" appears approximately normal but slightly right skewed.
- Most individuals have a salary in the range 60000 to 100000.
- The Median of Total salary is 80000.

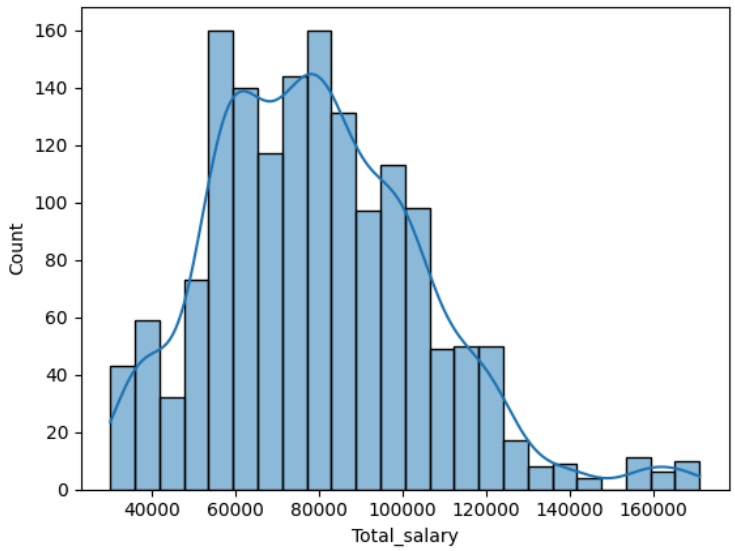

Figure 11: Histogram of Total_salary

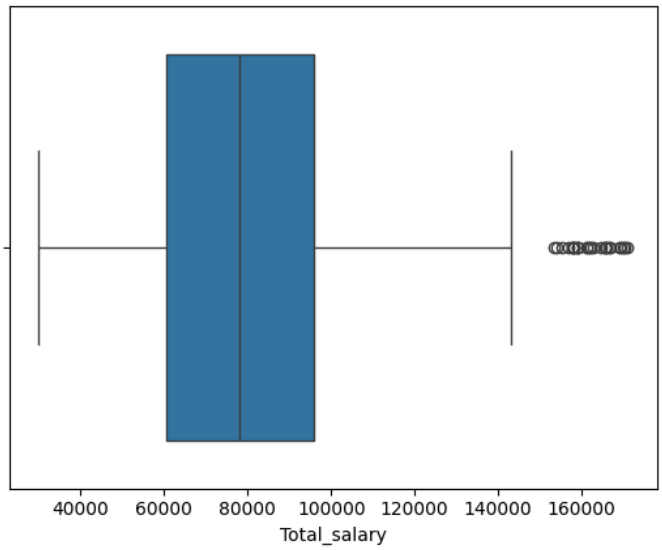

Figure 12: Barplot of Total_salary

Treating the outliers in Total_salary using clip()

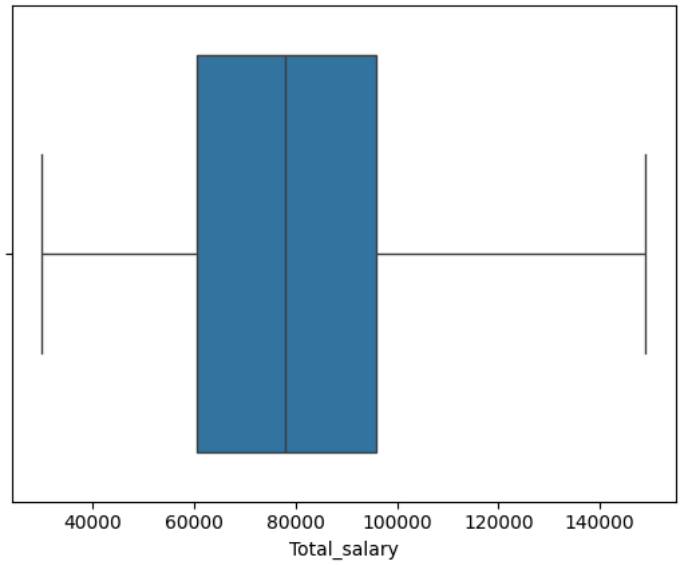

Figure 13: Bar Plot of Total_salary after treating outliers

Understanding the distribution of Gender and Profession

In the given data set, the number of Salaried Males and Females are more comparatively with one who have business.

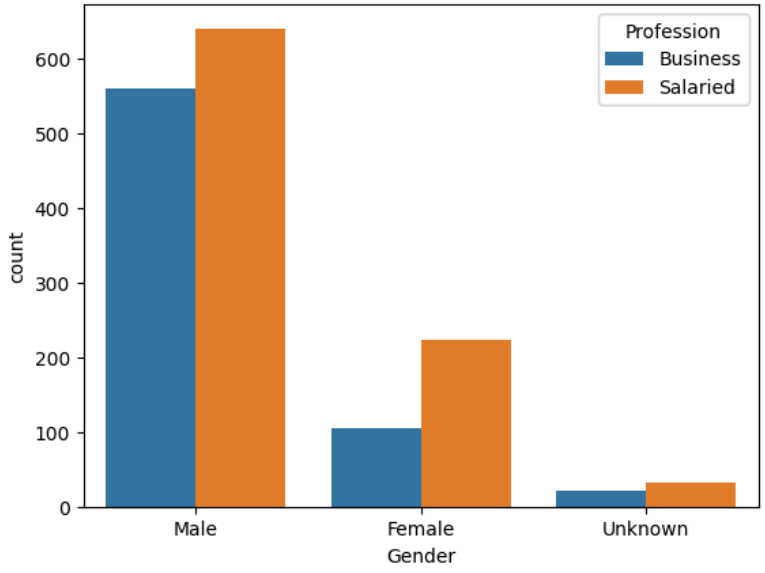

Figure 14: Count plot of Gender and Profession

Understanding the distribution between Make and Personal_loan

1. **SUV Owners:**
   - A significantly higher number of SUV owners do not have a personal loan compared to those who do.
   - This could imply that SUV owners may be more financially stable or less reliant on personal loans.
2. **Sedan Owners:**
   - More Sedan owners have a personal loan than not.
   - This might suggest that individuals buying sedans are more likely to take loans.
3. **Hatchback Owners:**
   - The counts are nearly equal for both loan takers and non-loan takers.
   - This could suggest that hatchback owners are a mixed group financially – some may prefer to not to take loans, while others rely on loans.

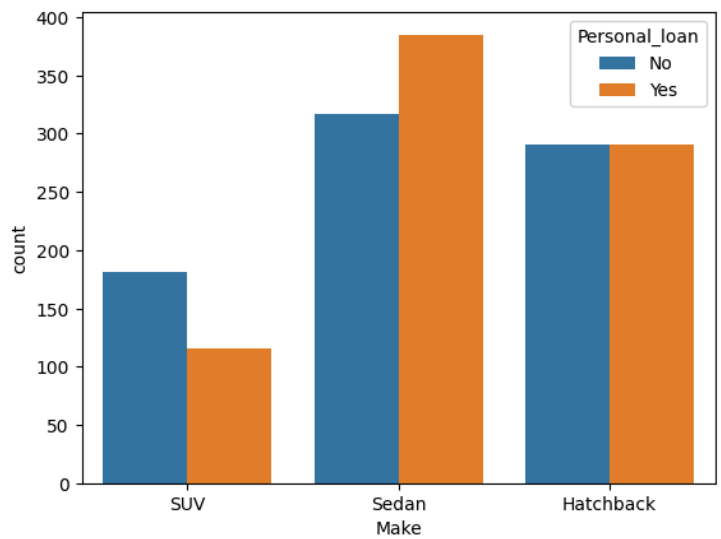

Figure 15: Countplot of Make and Personal_loan

Understanding the distribution of Make

- Sedan is the most popular type, with over 700 units.
- Hatchback comes second, with just under 600 units.
- SUV is the least common, with about 300 units.
-

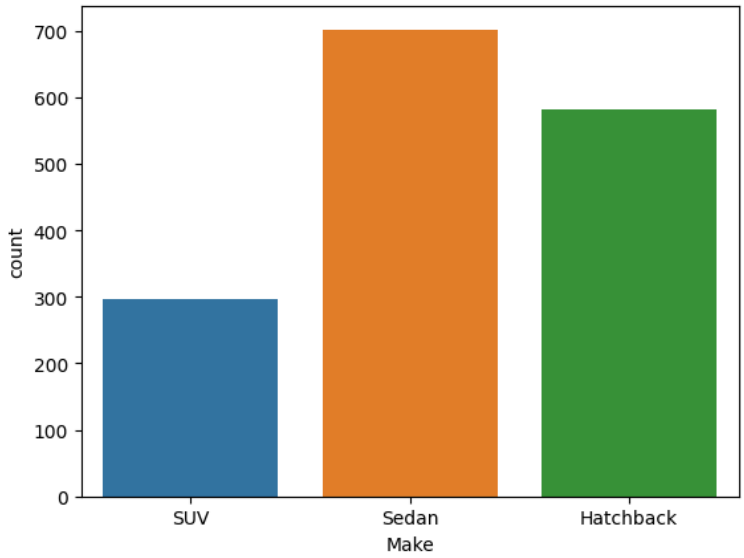

Figure 16: Countplot of Make

Bivariate Analysis

- **Visualising relationship between all numerical variables**

1. Salary vs. Total_Salary: Strong positive linear correlation.Higher Salary generally leads to higher Total_Salary.
2. Partner_Salary vs. Total_Salary: Also positively correlated, but more scattered.Many individuals have a Partner_Salary of 0 — indicating no partner or non-working partner.
3. Age vs. Salary: Weak to moderate positive trend — older people tend to earn more.
4. No_of_Dependents vs. Salary/Price: No clear linear relationship; data is scattered.
5. Total_Salary vs. Price: Positive relationship- higher total salary is associated with higher price Car.
6. Salary vs. Price: positive correlation — higher earners tend to be associated with higher price cars.

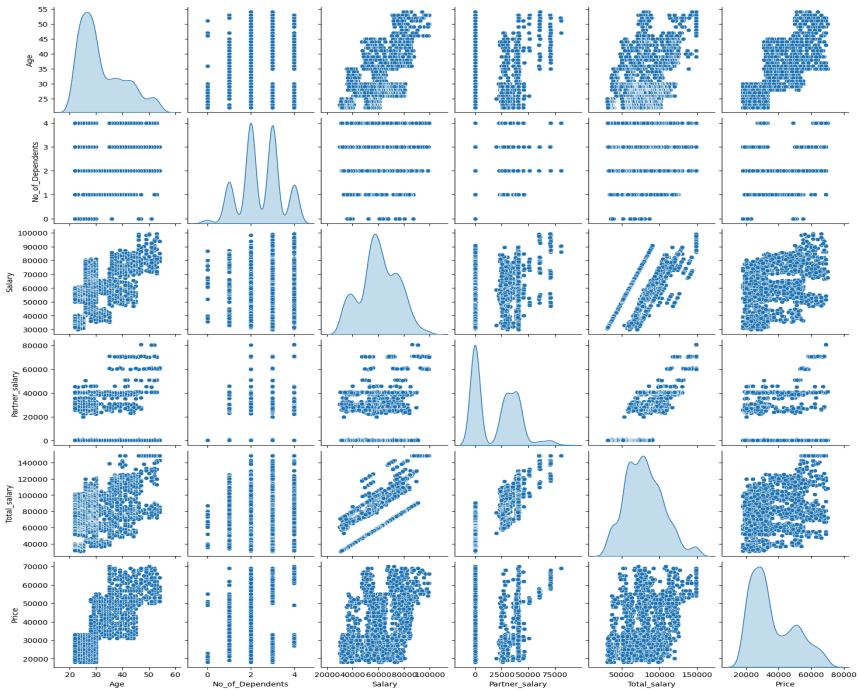

Figure 17: Pairplot for all numeric column

Visualising the correlation between all numerical variables

1. **Strong Positive Correlations**
   - Age and Price (0.80):Older individuals tend to be associated with higher price cars.
   - Salary and Total_salary (0.64).
   - Partner_salary and Total_salary (0.82).
     Total_salary is the sum of Salary and Partner_salary.
2. **Moderate Correlations**
   - Age and Salary (0.62):This suggests that salary tends to increase with age.
   - Partner_salary and Price (0.36):Indicates a moderate relationship between a partner's salary and the price.
3. **Low or No Significant Correlation**
   - No_of_Dependents with all other variables (max ~0.15):This variable does not have any significant correlation with other numerical features in the dataset.

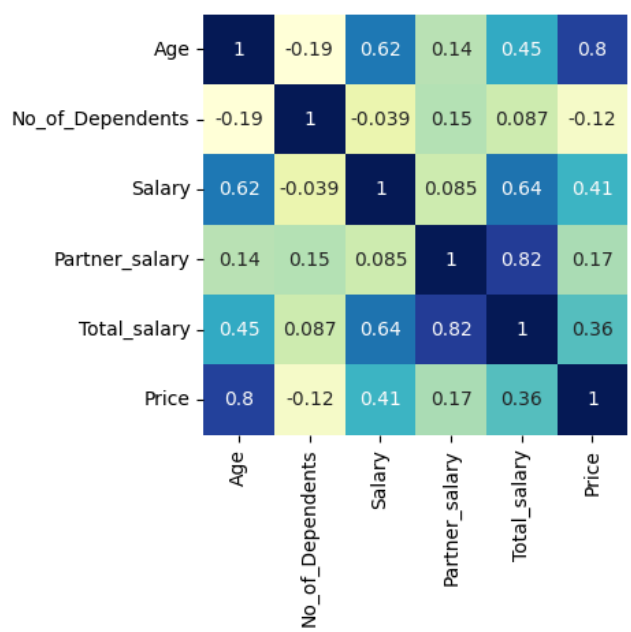

Figure 18: Correlation matrix all numeric column

Visualising the relationship between Age vs Gender

- In the given dataset an average age of females appears to be higher than that of males.
- This indicates that female individuals tend to be older than male.

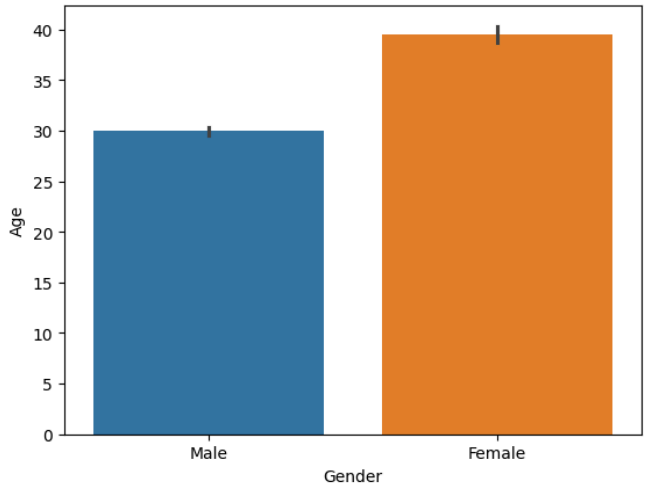

Figure 19: Age vs Gender

Visualising the relationship between Gender and Salary

- The average salary of females is higher than that of males.
- Females appear to earn approximately above 65,000, while males earn below 60,000.

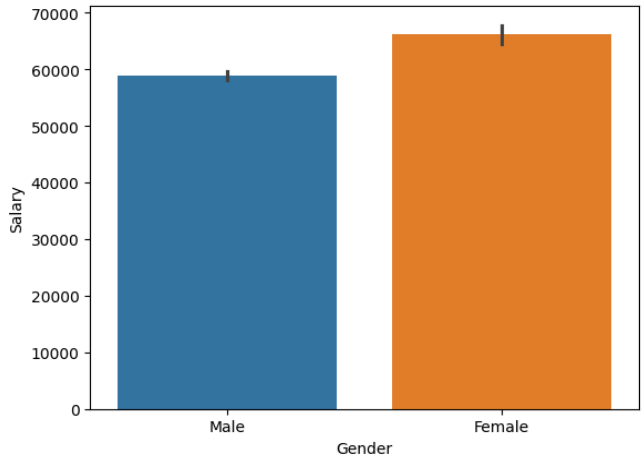

Figure 20: Gender vs Salary

Visualising the relationship between Education, Salary and Gender

1. **Education vs. Salary**
    - Post Graduates earn significantly higher salaries compared to Graduates.
    - The average salary for Post Graduates is well above 70,000.
    - The average salary for Graduates is below 55,000.
2. **Gender Comparison (within Education Levels)**
    - Post Graduate Level:
        1. Females earn the highest among all groups—above 70,000.
        2. Males earn slightly less—approximately 65,000.
        3. Individuals with Unknown gender earn slightly less than males but still around 65,000.
    - Graduate Level:
        1. Females again earn more than males—around 53,000.
        2. Males earn approximately 45,000.
        3. Unknown gender individuals earn slightly less than males—about 43, 000.

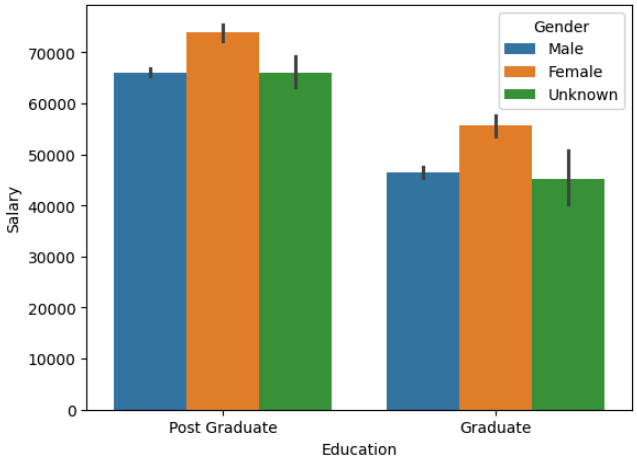

Figure 21: barplot of Education vs Salary Vs Gender

Visualising the relationship between Marital_status and Total_salary

- People who are married have more than 80000 as Total_salary.
- People who are not married have less than 60000 as Total_salary.

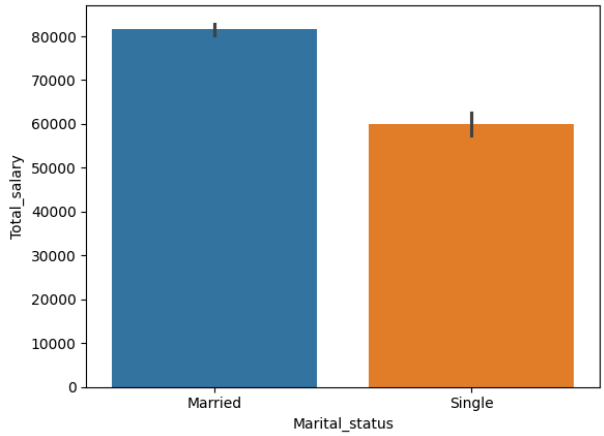

Figure 22: Bar Plot of Marital_status vs Total_salary

Visualising the relationship between Price and Make

- SUVs Are the Most Expensive price of SUVs are more than 55000.
- Sedans Are Mid-Range between SUV and Hatchback (more than 20000 and less than 35000).
- Hatchbacks Are the Most Affordable(less than 25000).

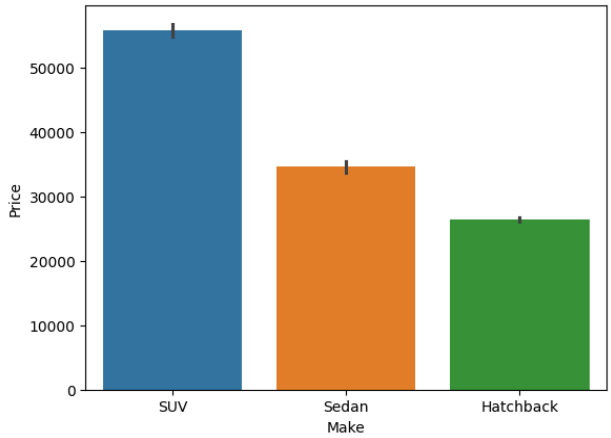

Figure 23: Barplot of Price vs Make

Visualizing the relationship between Age and Price with Gender as hue

1. **Price Increases with Age**
   - Across all gender categories, price generally increases with age, especially from age 30 to 50.
   - Suggests that older individuals tend to purchase more expensive cars.
2. **Gender Differences**
   - Male vs Female:
     1. Up to age ~45: both genders follow a similar trend, with some small variations.
     2. From age 45 to 50, females tend to spend slightly more than males, peaking higher.
     3. After 50, both slightly decline or stabilize.
   - Unknown Gender:
     1. Maintains significantly lower price levels until age 30, then a sharp rise.

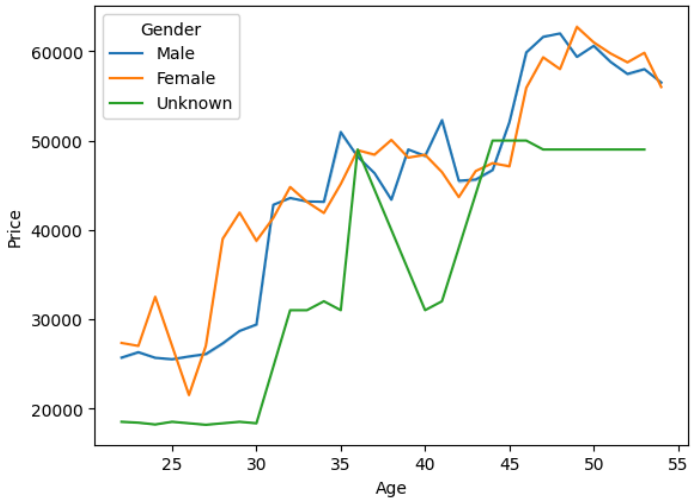

Figure 24: Preference on Gender by Age vs Price

Visualizing the relationship between Gender and Price with Make as hue

1. **SUVs:**
   - Males spend the most on SUVs (~$60,000).
   - Females spend slightly less (~$54,000).
   - The Unknown gender group pays the least (~$49,000).

     This suggests SUVs are most popular and most valued among males.
2. **Sedans:**
   - Females pay more for sedans ($43,000) compared to males ($34,000).
   - The price drops significantly for the Unknown group (~$20,000).

     Indicates that sedans are particularly preferred by female buyers,potentially due to a balance of comfort, size, and price.
3. **Hatchbacks:**
   - Females pay the most (~$28,000),
   - Followed by males (~$27,000),
   - Then Unknown (~$19,000).

     Pricing is more consistent across genders so it is considered as a budget-friendly car.

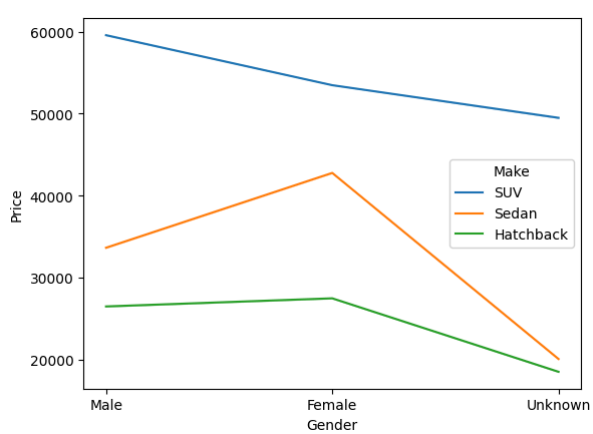

Figure 25: Preference on Make by Gender vs Price

Visualising the relationship between House_loan and Price

The individual who has the house loan tends to spend slightly less on cars than one who doesn't have a house loan.

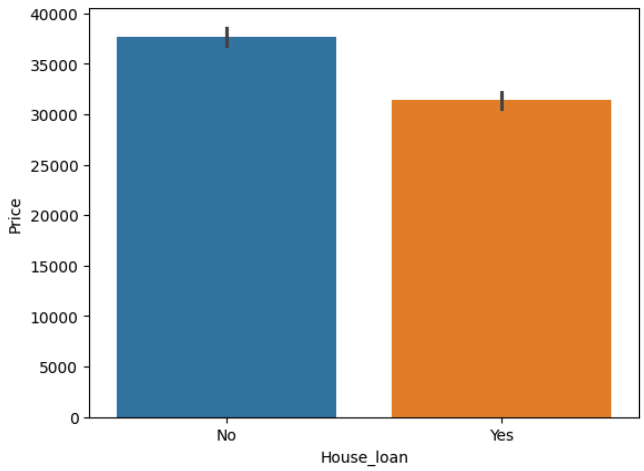

Figure 26: Barplot of House_loan and Price

Visualising the relationship between Personal_loan and Price

The individuals who have the personal loan tend to spend slightly less on cars than one who doesn't have a personal loan.

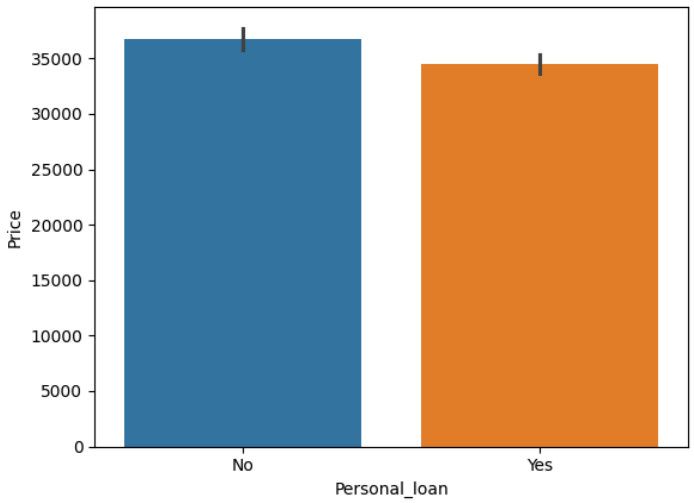

Figure 27: Bar Plot of Personal_loan vs price

Car Price vs House_loan Status with Personal_loan as hue

- Customers without a house loan:
  - Tend to buy more expensive cars(especially in the ₹40,000–₹60,000 range).
  - Those with personal loans (blue bars) are still active in purchasing higher price cars.
- Customers with a house loan:
  - Prefer lower to mid-range cars (₹20,000–₹35,000).
  - Very few spend above ₹40,000.

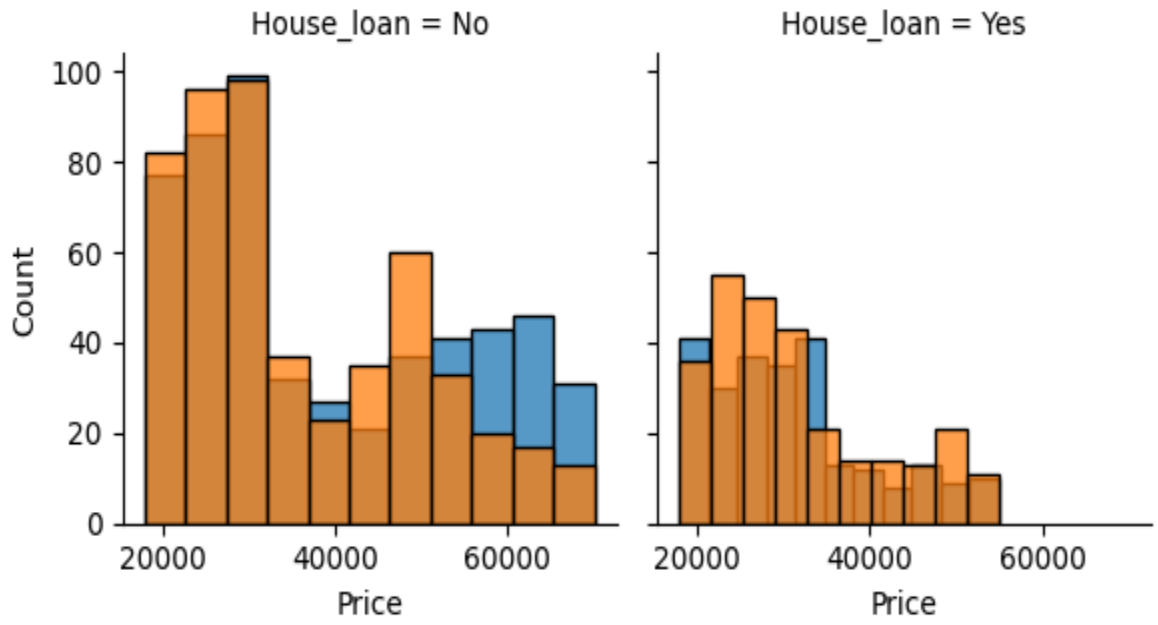

Figure 28: FacetGrid of Price vs House_loan

Car Price vs Personal_loan Status with House_loan as hue

- Customers without a personal loan:
  - A significant number purchase higher-end cars (₹40,000–₹65,000).
  - Most high spenders are not burdened with house loans (blue bars dominate the right side).
- Customers with a personal loan:
  - Generally stick to mid-range cars (₹20,000–₹40,000).
  - Having both personal and house loans is associated with minimal high-end car purchases.

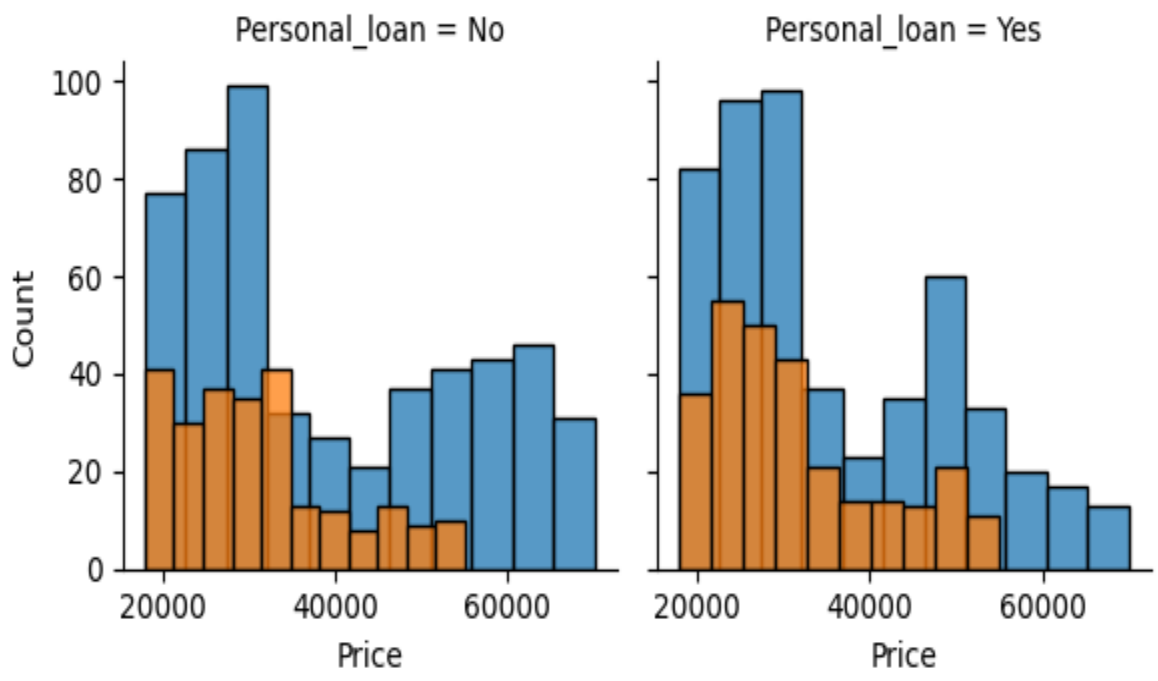

Figure 29: FacetGrid of Price vs Personal_loan

Q & A:

1.Do men tend to prefer SUVs more compared to women?

- No, from the above series it is clear that women prefer SUVs more compared to men.
- The count of women who prefer SUVs is 173 whereas the count of men who prefer SUVs is 118.

2. What is the likelihood of a salaried person buying a Sedan?

- Among 1528 total count of Profession column
    1. 864 are Salaried
        1. 370 are likely to buy Sedan.
        2. 291 are likely to buy a Hatchback.
        3. 203 are likely to buy an SUV.
    2. 664 are having Business
        1. 287 are likely to buy a Sedan.
        2. 289 are likely to buy a Hatchback.
        3. 88 are likely to buy an SUV.
- 42.82% of salaried people are likely to buy a Sedan.

3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

The data doesn't support Sheldon Cooper's claim. In fact, salaried males are over three times more likely to buy a Sedan than an SUV, making them a harder target for SUV sales.

4. How does the amount spent on purchasing automobiles vary by gender?

- Females spent more on purchasing automobiles compared to Male.
- The average price spent by females is 47705.16.
- The average price spent by Male is 32817.34.

5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

- An amount of 26644000 was spent on purchasing automobiles by individuals who took a personal loan.
- An amount of 28399000 was spent on purchasing automobiles by individuals who did not take a personal loan.

6. How does having a working partner influence the purchase of higher-priced cars?

- An average of 35730.31 amount spent to purchase the automobile who have working partner, which is more than the amount spent to purchase the automobile who don't have working partner.
- This indicates that a working partner influences the purchase of higher-priced cars.

7. Who tends to take more Personal loan: salaried or business holders?

- The business holders tend to take less personal loan compared to salaried individuals.
- Among 1581 observations:
    - 685 are business holders: 363 take loans and 322 don't take.
    - 896 are salaried individuals: 429 take loan and 467 don't take.

8. Who tends to take more House loans: salaried or business holders?

- The business holders tend to take less house loans compared to salaried individuals.
- Among 1581 observations:

    - 685 are business holders: 229 take loan and 456 don't take.
    - 896 are salaried individuals: 298 take loan and 598 don't take.

9. Who tends to take more Personal loan: Graduates or Post Graduates?

- The Graduates tend to take less personal loan compared to Postgraduate individuals.
- Among 1581 observations:
    - 596 are Graduates: 308 take loan and 288 don't take.
    - 985 are Post Graduates: 484 take loan and 501 doesn't take.

10. Who tends to take more House loans: Graduates or Post Graduates?

- The Graduates tend to take less house loan compared to Postgraduate individuals.
- Among 1581 observations:
    - 596 are Graduates: 202 takes loan and 394 doesn't take.
    - 985 are Post Graduates: 325 takes loan and 660 doesn't take

Actionable Insights and Recommendations

Insights

- Most buyers are young (avg. age 32), salaried, married males with a postgraduate degree.
- Sedans are the top car choice, followed by SUVs and hatchbacks.
- Cars are typically priced between ₹25,000 and ₹50,000, with few high-end purchases.
- Older age and higher income strongly correlate with buying more expensive cars.
- Dual-income households (with working partners) have significantly higher total income.
- Families with more dependents tend to spend slightly less on cars.
- The individuals with Postgraduate and Salaried are more likely to use personal loans.
- The Postgraduates are more likely to use house loans.

Actionable Recommendations

- Married postgraduates in the 30–40 age group are high-value targets because they show the strongest purchase behavior.
- Sedans are the most popular type of car among customers—keep them in stock and promote them.
- Prioritize marketing to dual-income families because they have significantly more purchasing power.
- Launch targeted promotions for mid-range cars (₹25,000–₹35,000) where most customers are buying—boost volume here.
- Show different car models and offers depending on income, loan status, and age.
- Suggest higher-end cars to older and higher-earning buyers; offer budget options to younger, low-income customers.
- Provide Offers like free maintenance, extended warranty, or child safety features for customers with multiple dependents.
- Prioritize marketing to postgraduates—they are more likely to take loans and make purchase decisions.
- Prioritize marketing to those who don't have house or personal loans because they are more likely to buy higher-priced cars.